

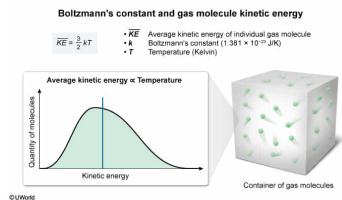
A large number of monoatomic ideal gas molecules are stored within a closed container at a temperature of 200 K. What is the average translational kinetic energy of the individual gas molecules? (Note: Use Boltzmann's constant  $k = 1.381 \times 10^{-23} \text{ J/K}$ .)

- ☐ A.  $2.8 \times 10^{-23} \text{ J}$  [5%]  
☐ B.  $4.2 \times 10^{-23} \text{ J}$  [8%]  
☐ C.  $2.8 \times 10^{-21} \text{ J}$  [57%]  
☒ D.  $4.2 \times 10^{-21} \text{ J}$  [27%]

Correct

28% answered correctly

#### Explanation:



The **kinetic molecular model** describes **macroscopic characteristics** of gases that are related to the constant motion of freely moving gas molecules. The kinetic molecular model assumes that a rigid, closed container contains a large number  $N$  of monoatomic **ideal gas molecules**. The total kinetic energy ( $K_{tr}$ ) of all gas molecules is related to the mass ( $m$ ) of each molecule and the average value of velocity squared ( $\overline{v^2}$ ):

$$K_{tr} = N \cdot \frac{1}{2} m \overline{v^2}$$

Applying the kinetic molecular model to a rigid cubic container allows the **pressure ( $P$ )** of the gas to be stated in terms of the total kinetic energy of all gas molecules and the container volume ( $V$ ):

$$P = \frac{2}{3} \cdot \frac{N \left( \frac{1}{2} m \overline{v^2} \right)}{V}$$

$$\frac{P}{V} = \frac{2}{3} \cdot \frac{N \left( \frac{1}{2} m \overline{v^2} \right)}{V}$$

Furthermore, Boltzmann's constant ( $k = 1.381 \times 10^{-23} \text{ J/K}$ ) may be used to restate the **ideal gas law** in terms of the behavior of individual gas molecules as:

$$P V = N k T$$

Combining the kinetic molecular model's expressions for ideal gas behavior and pressure produces an expression for the **average kinetic energy ( $KE$ )** of an individual gas molecule at a given temperature in **Kelvin (K)**:

$$\frac{P}{V} = \frac{2}{3} \cdot \frac{N}{V}$$

$$\frac{2}{3} \overline{KE} = k T$$

$$\overline{KE} = \frac{3}{2} k T$$

In this question, ideal gas molecules are stored within a fixed container at a temperature of 200 K. Therefore, the average translational kinetic energy of an individual monoatomic ideal gas molecule is:

$$\overline{KE} = \frac{3}{2} k T$$

$$\overline{KE} = \left( \frac{3}{2} \right) (1.381 \times 10^{-23} \text{ J/K}) (200 \text{ K})$$

$$\overline{KE} = 4.2 \times 10^{-21} \text{ J}$$

Therefore, the average kinetic energy of an ideal gas molecule stored within a closed container at 200 K is  $4.2 \times 10^{-21} \text{ J}$  (**Choice D**).

#### Educational objective:

The kinetic molecular model uses the behavior of individual gas molecules to explain the macroscopic characteristics of gases. Boltzmann's constant can be used to determine the average translational kinetic energy of a single ideal gas molecule at a given temperature.

Time spent: 0 sec  
 Subject:  
 Physics

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 System:  
 Fluids